

HW17 - Pico Encoder, Load Sensing, and Computer Graphics

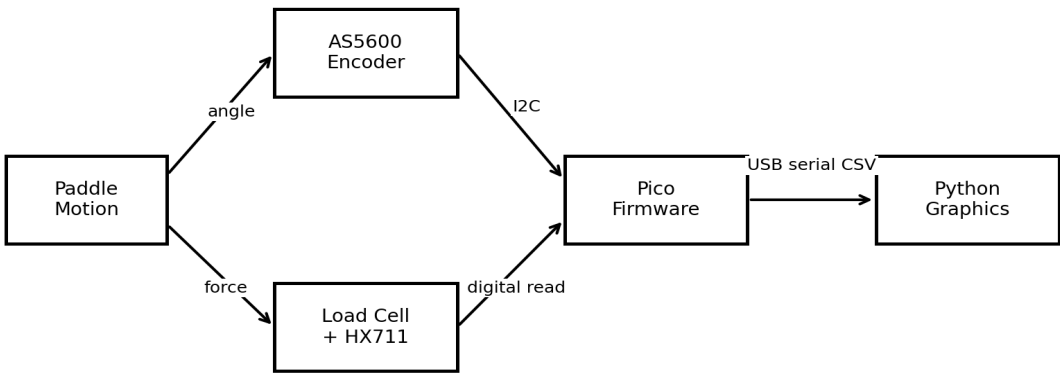
Junzhe Lei

Assignment Goal

The goal of this assignment is to read the AS5600 magnetic encoder and the load cell, stream the paddle position and force data to the computer, and update a real-time graphics display. The AS5600 is read over I2C at address 0x36. The load cell is read through an HX711 amplifier.

System Overview

HW17 System Block Diagram

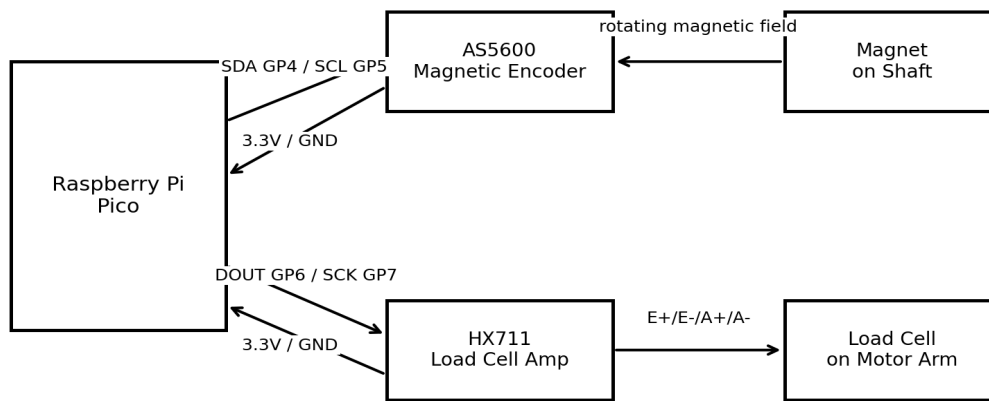


Goal: read position and force, stream both measurements, and update real-time computer graphics.

Wiring

Connection	Pico Pin / Note
AS5600 SDA	GP4
AS5600 SCL	GP5
AS5600 VCC/GND	3.3V / GND
HX711 DOUT	GP6
HX711 SCK	GP7
HX711 VCC/GND	3.3V / GND
USB serial	CSV data to Python graphics

HW17 Circuit / Wiring Diagram

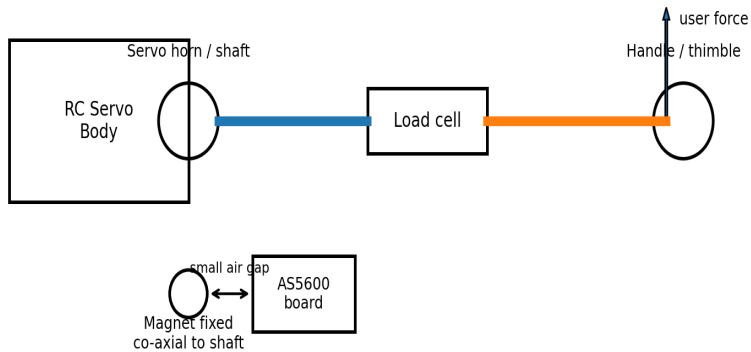


USB serial CSV to computer: time_ms, angle_raw, angle_deg, load_raw, force_N

Mechanical Layout

The magnet is mounted coaxially with the servo shaft so the magnetic field rotates over the AS5600 chip. The AS5600 board stays fixed with a small air gap to the magnet. The load cell is placed in the arm so the tangential force applied at the handle is sensed.

Mechanical Sketch: Encoder Magnet and Load Cell Placement



The magnet rotates with the servo shaft while the AS5600 board stays fixed. The load cell is mounted in series with the arm so tangential user force is measured.

Firmware and Data Stream

The Pico firmware initializes I2C, reads the AS5600 angle registers 0x0E and 0x0F, masks the result to 12 bits, reads the HX711 load cell amplifier, tares the load cell at startup, and prints one CSV row about every 20 ms.

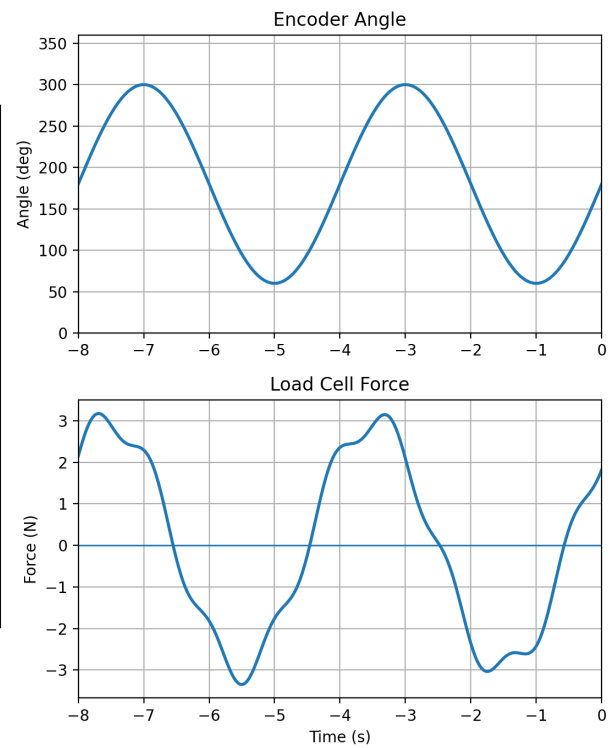
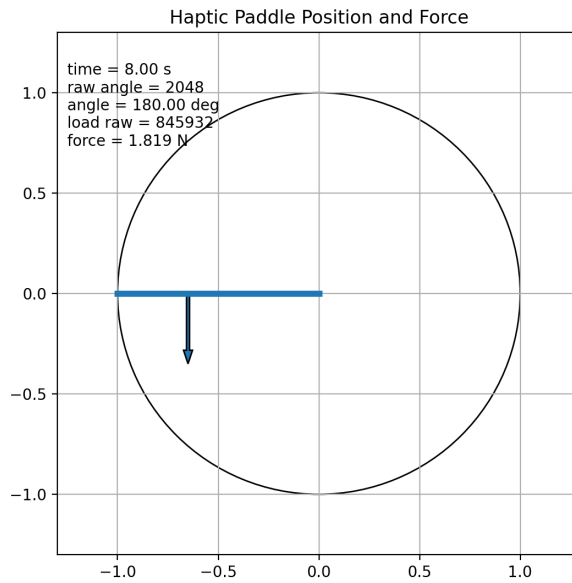
CSV format: time_ms,angle_raw,angle_deg,load_raw,force_N

Angle equation: $\text{angle_deg} = \text{angle_raw} * 360 / 4096$

Force equation: $\text{force_N} = (\text{load_raw} - \text{tare_offset}) / \text{counts_per_newton}$

Computer Graphics

The Python script reads the CSV serial data and updates three graphics: a paddle arm angle display, an encoder angle history plot, and a load-cell force history plot. The force arrow is drawn perpendicular to the paddle to show the direction and approximate magnitude of the measured tangential force.



Notes

The calibration constant in main.c should be adjusted with a known weight for accurate force units. The provided version still streams raw load-cell counts, so the data can be replotted or recalibrated later.